

# Byung-Kwan Lee

Ph.D., School of Electrical Engineering at KAIST

Research Scientist at  NVIDIA

 bklee.nano@gmail.com ·  byungkwanl@nvidia.com ·  Homepage

 LinkedIn ·  Github ·  Google Scholar ·  Huggingface

## Research Interest

Building efficient, high-performing Vision-Language Models (VLMs), with focus on:

- Distillation
- RL
- Agent
- Pruning

## Work Experience

### NVIDIA • Research Scientist

Oct. 2025 - Current

**LEAD ZPPO: Zone of Proximal Policy Optimization** [COMPLETED] RL post-training that keeps the teacher inside the prompt rather than the policy gradient, using BCQ/NCQ question reformulations and a prompt replay buffer to lift small students on hard questions without policy drift.

Under Review U.S. Provisional Patent Filed 2026

**LEAD AXPO: Agent eXplorative Policy Optimization** [COMPLETED] Agentic RL training that recovers tool usage through tool-call resampling, improving multimodal reasoning performance against larger baselines.

Under Review U.S. Provisional Patent Filed 2026 Internal Tech Transfer

**LEAD Masking Teacher and Reinforcing Student** [COMPLETED] Mask-progressive RL distillation that gradually un.masks teacher weights and uses offline RL with accuracy & distillation rewards.

CVPR 2026 Accept U.S. Non-Provisional Patent Filed 2026

**LEAD Unified RL & Imitation Learning for VLMs** [COMPLETED] Combines RL with adversarial imitation, using an LLM-based discriminator and multi-teacher guidance to build lightweight yet powerful VLMs.

NeurIPS 2025 Accept U.S. Patent Upgraded to Non-Provisional 2025

**LEAD GenRecal** [COMPLETED] Cross-architecture VLM distillation via a Recalibrator that aligns heterogeneous token representations regardless of vocabulary, token splits, or index ordering.

ECCV 2026 Accept U.S. Patent Upgraded to Non-Provisional 2025 Internal Tech Transfer

**LEAD VLsI: Verbalized Layers-to-Interactions** [COMPLETED] Hardened the verbalizer-based layer-wise distillation into a production-ready recipe and led its internal tech transfer into NVIDIA's VLM development pipeline.

CVPR 2025 Accept U.S. Patent Upgraded to Non-Provisional 2025 Internal Tech Transfer

### NVIDIA • Research Intern

Oct. 2024 - Oct. 2025

**LEAD Unified RL & Imitation Learning for VLMs** [INITIATED] First formulation of the RL + adversarial imitation training pipeline; initial experiments and team setup.

U.S. Provisional Patent Filed 2025

**LEAD GenRecal** [INITIATED] Initial design of the Recalibrator framework for cross-tokenizer VLM distillation; first proof-of-concept and internal demo.

U.S. Provisional Patent Filed 2025

**LEAD VLsI: Verbalized Layers-to-Interactions** [INITIATED] Layer-wise distillation using intermediate verbalizers, enabling small VLMs (2B/7B) to align with large VLMs' reasoning progression and outperform GPT-4V.

CVPR 2025 Accept U.S. Provisional Patent Filed 2025

## Education

### KAIST

Ph.D., School of Electrical Engineering

◦ GPA: 3.77/4.3

◦ Dissertation: *Building High-performing, Efficient-size Vision Language Models: Merge, Modify, and Distill* [Link]

Mar. 2020 - Aug. 2025

### KAIST

M.S., The Cho Chun Sik Graduate School of Green Transportation

◦ GPA: 3.72/4.3

◦ Thesis: *Training Encoder-Attention through Fully-Connected CRFs for Efficient End-to-End Lane Detection Model* [Link]

Mar. 2018 - Feb. 2020

### Hanyang University

B.S., Mathematics and Electronic Engineering

◦ GPA: 3.86/4.5

◦ Thesis: *Learning Rank of Evaluated Songs for Classifying Unknown Music by Convolutional Neural Network* [Link]

Mar. 2014 - Feb. 2018

## Publications

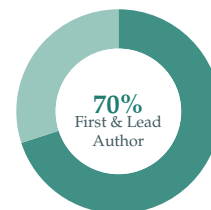
**Overall** 18 Accepts · 4 Pending · 5 Tech Reports

**CV** 5 CVPR · 2 ICCV · 3 ECCV · 1 ICIP

**ML** 3 NeurIPS · 1 ICLR

**NLP** 1 ACL · 1 EMNLP

**Journal** 1 Pattern Recognition



■ 70% (19/27) First & Lead Author  
■ 30% (8/27) Co-Author

### 1. Zone of Proximal Policy Optimization: Teacher in Prompts, Not Gradients

Byung-Kwan Lee, Ximing Lu, Shizhe Diao, Minki Kang, Saurav Muralidharan, Karan Sapra, Andrew Tao, Pavlo Molchanov, Yejin Choi, Yu-Chiang Frank Wang, Ryo Hachiuma

Under Review [Paper][Project]

U.S. Patent Application Filed (Provisional), NVIDIA Research, 2026

### 2. SpatialClaw: Rethinking Action Interface for Agentic Spatial Reasoning

Seokju Cho, Ryo Hachiuma, Abhishek Badki, Hang Su, Byung-Kwan Lee, Chan Hee Song, Sifei Liu, Subhashree Radhakrishnan, Seungryong Kim, Yu-Chiang Frank Wang, Min-Hung Chen

Under Review [Paper][Project][Code]

U.S. Patent Application Filed (Provisional), NVIDIA Research, 2026

### 3. Agent eXplorative Policy Optimization for Multimodal Agentic Reasoning

Minki Kang, Shizhe Diao, Ryo Hachiuma, Sung Ju Hwang, Pavlo Molchanov, Yu-Chiang Frank Wang, Byung-Kwan Lee

Under Review [Paper][Project]

U.S. Patent Application Filed (Provisional), NVIDIA Research, 2026

### 4. Hide to See: Reasoning-prefix Masking for Visual-anchored Thinking in VLM Distillation

Seonghoon Yu, Dongjun Nam, Byung-Kwan Lee<sup>†</sup>, Jeany Son<sup>†</sup>

Under Review [Paper][Code]

### 5. Why and When Visual Token Pruning Fails? A Study on Relevant Visual Information Shift in MLLMs Decoding

Jiwan Kim, Kibum Kim, Wonjoong Kim, Byung-Kwan Lee, Chanyoung Park

European Conference on Computer Vision (ECCV), 2026 [Paper][Project]

### 6. GenRecal: Generation after Recalibration from Large to Small Vision Language Models

Byung-Kwan Lee, Ryo Hachiuma, Yong Man Ro, Yu-Chiang Frank Wang, and Yueh-Hua Wu

European Conference on Computer Vision (ECCV), 2026 [Paper][Project]

U.S. Patent Application Filed (Non-Provisional), NVIDIA Research, 2025

- 7. Masking Teacher and Reinforcing Student for Distilling Vision-Language Models**  
Byung-Kwan Lee, Yu-Chiang Frank Wang, Ryo Hachiuma  
IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), 2026 [[Paper](#)]  
U.S. Patent Application Filed (Non-Provisional), NVIDIA Research, 2026
- 8. Recursive Think-Answer Process for LLMs and VLMs**  
Byung-Kwan Lee<sup>\*</sup>, Youngchae Chee<sup>\*</sup>, Yong Man Ro  
IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR) Findings, 2026 [[Paper](#)][[Project](#)]
- 9. RefineBench: Evaluating Refinement Capability in Language Models**  
Young-Jun Lee<sup>\*</sup>, Seungone Kim<sup>\*</sup>, Byung-Kwan Lee, Minkyong Moon, Yechan Hwang, Jong Myoung Kim, Graham Neubig, Sean Welleck, Ho-Jin Choi  
International Conference on Learning Representations (ICLR), 2026 [[Paper](#)][[Project](#)]  
Best Runner-Up Award, Workshop for Multi-Turn Interactions in Large Language Models, NeurIPS 2025 (Oral, Top 1%) [[Link](#)]
- 10. Unified Reinforcement and Imitation Learning for Vision-Language Models**  
Byung-Kwan Lee, Ryo Hachiuma, Yong Man Ro, Yu-Chiang Frank Wang, Yueh-Hua Wu  
Neural Information Processing Systems (NeurIPS), 2025 [[Paper](#)][[Project](#)]  
U.S. Patent Application Filed (Non-Provisional), NVIDIA Research, 2025
- 11. MultiVerse: A Multi-Turn Conversation Benchmark for Evaluating Large Vision and Language Models**  
Young-Jun Lee, Byung-Kwan Lee, Jianshu Zhang, Yechan Hwang, Byungsoo Ko, Han-Gyu Kim, Dongyu Yao, Xuankun Rong, Eojin Joo, Seung-Ho Han, Bowon Ko, Ho-Jin Choi  
IEEE/CVF International Conference on Computer Vision (ICCV), 2025 [[Paper](#)][[Project](#)]  
Workshop for Knowledge-Intensive Multimodal Reasoning, ICCV 2025 [[Link](#)]
- 12. Enhancing Conversational Agents with Skill-of-Mind-Infused Large Language Model**  
Young-Jun Lee, Byung-Kwan Lee, Dokyong Lee, Junyoung Youn, Kyeong-Jin Oh, Yechan Hwang, Ho-Jin Choi  
Technical Report [[Paper](#)][[Code](#)][[Huggingface Model](#)]
- 13. VLsI: Verbalized Layers-to-Interactions from Large to Small Vision Language Models**  
Byung-Kwan Lee, Ryo Hachiuma, Yu-Chiang Frank Wang, Yong Man Ro<sup>†</sup>, Yueh-Hua Wu<sup>†</sup>  
IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), 2025 [[Paper](#)][[Project](#)]  
U.S. Patent Application Filed (Non-Provisional), NVIDIA Research, 2025
- 14. Are Vision-Language Models Truly Understanding Multi-vision Sensor?**  
Sangyun Chung<sup>\*</sup>, Youngjoon Yu<sup>\*</sup>, Youngchae Chee, Se Yeon Kim, Byung-Kwan Lee, Yong Man Ro  
Technical Report [[Paper](#)][[Code](#)]
- 15. Phantom of Latent for Large Language and Vision Models**  
Byung-Kwan Lee, Sangyun Chung, Chae Won Kim, Beomchan Park, Yong Man Ro  
Technical Report [[Paper](#)][[Code](#)][[Huggingface Model](#)]
- 16. SPARK: Multi-Vision Sensor Perception and Reasoning Benchmark for Large-scale Vision-Language Models**  
Youngjoon Yu<sup>\*</sup>, Sangyun Chung<sup>\*</sup>, Byung-Kwan Lee, Yong Man Ro  
Technical Report [[Paper](#)][[Code](#)][[Huggingface Dataset](#)]
- 17. TroL: Traversal of Layers for Large Language and Vision Models**  
Byung-Kwan Lee, Sangyun Chung, Chae Won Kim, Beomchan Park, Yong Man Ro  
Empirical Methods in Natural Language Processing (EMNLP), 2024 [[Paper](#)][[Code](#)][[Huggingface Model](#)]
- 18. Meteor: Mamba-based Traversal of Rationale for Large Language and Vision Models**  
Byung-Kwan Lee, Chae Won Kim, Beomchan Park, Yong Man Ro  
Neural Information Processing Systems (NeurIPS), 2024 [[Paper](#)][[Code](#)][[Huggingface Model](#)]  
31st Samsung HumanTech Paper Awards in section of Computer Science & Engineering

19. **MoAI: Mixture of All Intelligence for Large Language and Vision Models**  
 Byung-Kwan Lee, Beomchan Park, Chae Won Kim, Yong Man Ro  
 European Conference on Computer Vision (ECCV), 2024, [[Paper](#)][[Code](#)][[Huggingface Model](#)]
20. **CoLLaVO: Crayon Large Language and Vision mOdel**  
 Byung-Kwan Lee, Beomchan Park, Chae Won Kim, Yong Man Ro  
 Findings of the Association for Computational Linguistics (ACL), 2024, [[Paper](#)][[Code](#)][[Huggingface Model](#)]  
 2024 KCC XAI Workshop, Best Paper Awards
21. **Causal Unsupervised Semantic Segmentation**  
 Junho Kim\*, [Byung-Kwan Lee\\*](#), and Yong Man Ro  
 Journal of Pattern Recognition [[Paper](#)][[Code](#)]
22. **Mitigating Adversarial Vulnerability through Causal Parameter Estimation by Adversarial Double Machine Learning**  
 Byung-Kwan Lee\*, Junho Kim\*, and Yong Man Ro  
 IEEE/CVF International Conference on Computer Vision (ICCV), 2023 [[Paper](#)][[Code](#)]
23. **Mitigating Dataset Bias in Image Captioning through CLIP Confounder-free Captioning Network**  
 YeonJu Kim, Junho Kim, [Byung-Kwan Lee](#), Sebin Shin, and Yong Man Ro  
 IEEE International Conference on Image Processing (ICIP), 2023 [[Paper](#)][[Code](#)]
24. **Demystifying Causal Features on Adversarial Examples and Causal Inoculation for Robust Network by Adversarial Instrumental Variable Regression**  
 Junho Kim\*, [Byung-Kwan Lee\\*](#), and Yong Man Ro  
 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), 2023 [[Paper](#)][[Code](#)]
25. **Masking Adversarial Damage: Finding Adversarial Saliency for Robust and Sparse Network**  
 Byung-Kwan Lee\*, Junho Kim\*, and Yong Man Ro  
 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), 2022 [[Paper](#)][[Code](#)]
26. **Distilling Robust and Non-Robust Features in Adversarial Examples by Information Bottleneck**  
 Junho Kim\*, [Byung-Kwan Lee\\*](#), and Yong Man Ro  
 Neural Information Processing Systems (NeurIPS), 2021 [[Paper](#)][[Code](#)]
27. **Towards Adversarial Robustness of Bayesian Neural Network through Hierarchical Variational Inference**  
 Byung-Kwan Lee\*, Youngjun Yu, and Yong Man Ro  
 Technical Report [[Paper](#)][[Code](#)]



## Reviewer Experience

|                                                                        |            |
|------------------------------------------------------------------------|------------|
| IEEE Transactions on Pattern Analysis and Machine Intelligence (TPAMI) | Journal    |
| IEEE Transactions on Neural Networks and Learning Systems (TNNLS)      | Journal    |
| IEEE Transactions on Circuits and Systems for Video Technology (TCSVT) | Journal    |
| IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)  | Conference |
| IEEE/CVF International Conference on Computer Vision (ICCV)            | Conference |
| European Conference on Computer Vision (ECCV)                          | Conference |
| International Conference on Learning Representations (ICLR)            | Conference |
| Neural Information Processing Systems (NeurIPS)                        | Conference |
| International Conference on Machine Learning (ICML)                    | Conference |
| AAAI Conference on Artificial Intelligence (AAAI)                      | Conference |
| Association for Computational Linguistic (ACL)                         | Conference |
| Empirical Methods in Natural Language Processing (EMNLP)               | Conference |

## Invited Talks and Awards

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|-----------------------------------------------------------------------------------------------------------------------------------------|-----------------------|
| Silver Reviewer Award, International Conference on Machine Learning (ICML)                                                              | 2026                  |
| Best Runner-Up Award (Oral, Top 1%), Multi-Turn Interactions in LLMs Workshop @ NeurIPS                                                 | 2025                  |
| Invited Talk in Kongju University                                                                                                       | 2025                  |
| Invited Talk in Hanyang University ERICA                                                                                                | 2025                  |
| Invited Talk in Kookmin University                                                                                                      | 2025                  |
| Invited Talk in Dongseo University                                                                                                      | 2025                  |
| 31st Samsung HumanTech Paper Awards in section of Computer Science & Engineering                                                        | 2025                  |
| KCC XAI Workshop, Best Paper Awards                                                                                                     | 2024                  |
| Invited Talk in Korea Institute of Science and Technology Information (KISTI)                                                           | 2024                  |
| Invited Talk in NAVER HyperCLOVA X                                                                                                      | 2024                  |
| Invited Talk in Kakao Brain                                                                                                             | 2024                  |
| 1st Award for Research Presentation in Center for Applied Research in Artificial Intelligence (CARAI)                                   | 2023                  |
| GIRE Research Mentor, Seoul International School – “MUSE”, Journal of Student Research [ <a href="#">Publication</a> ]                  | 2023                  |
| KAIST SW IT Academy, AI Project Director of 1st Award Topic                                                                             | 2023                  |
| Invitation to Presenter in Korean Conference on Computer Vision (KCCV)                                                                  | 2023                  |
| KAIST SW IT Academy, Lecturer of Data Structure & Algorithm, Deep Learning, Computer Vision                                             | 2023                  |
| Invitation to Presenter in Korean Conference on Computer Vision (KCCV)                                                                  | 2022                  |
| 2nd Award for Research Presentation in Center for Applied Research in Artificial Intelligence (CARAI)                                   | 2022                  |
| KAIST Fellowship                                                                                                                        | Mar. 2020 - Feb. 2024 |
| National Government Fellowship                                                                                                          | Mar. 2018 - Feb. 2020 |
| 1st President’s Award for International Autonomous Vehicle Competition, Team Leader [ <a href="#">YouTube</a> ][ <a href="#">Talk</a> ] | 2018                  |